

183 Final Project R code

2026-05-05

```
# install.packages("readxl")
# install.packages("ggplot2")
# install.packages("dplyr")
# install.packages("car")
library(readxl)
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(car)
```

```
## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##   recode
```

```
# Set working directory if needed
# setwd("your/path/here")

# Read Excel file
temp <- read.csv("temp.csv")
ph <- read.csv("ph.csv")
temp_data <- temp[0:4]
ph_data <- ph[0:4]

# Ensure 'trial' and the treatment variables are factors for ANOVA
temp_data$Trial <- as.numeric(temp_data$Trial)
```

```
temp_data$Temp <- as.numeric(temp_data$Temp)

ph_data$Trial <- as.numeric(ph_data$Trial)
ph_data$pH <- as.numeric(ph_data$pH)

names(temp_data)[names(temp_data) == "Diameter..cm."] <- "Diameter"
names(ph_data)[names(ph_data) == "Diameter..cm."] <- "Diameter"

head(temp_data)
```

```
##   Temp Trial Rep Diameter
## 1   11    1  1     0.0
## 2   11    1  2     0.0
## 3   11    1  3     0.0
## 4   11    1  4     0.0
## 5   15    1  1     3.1
## 6   15    1  2     3.7
```

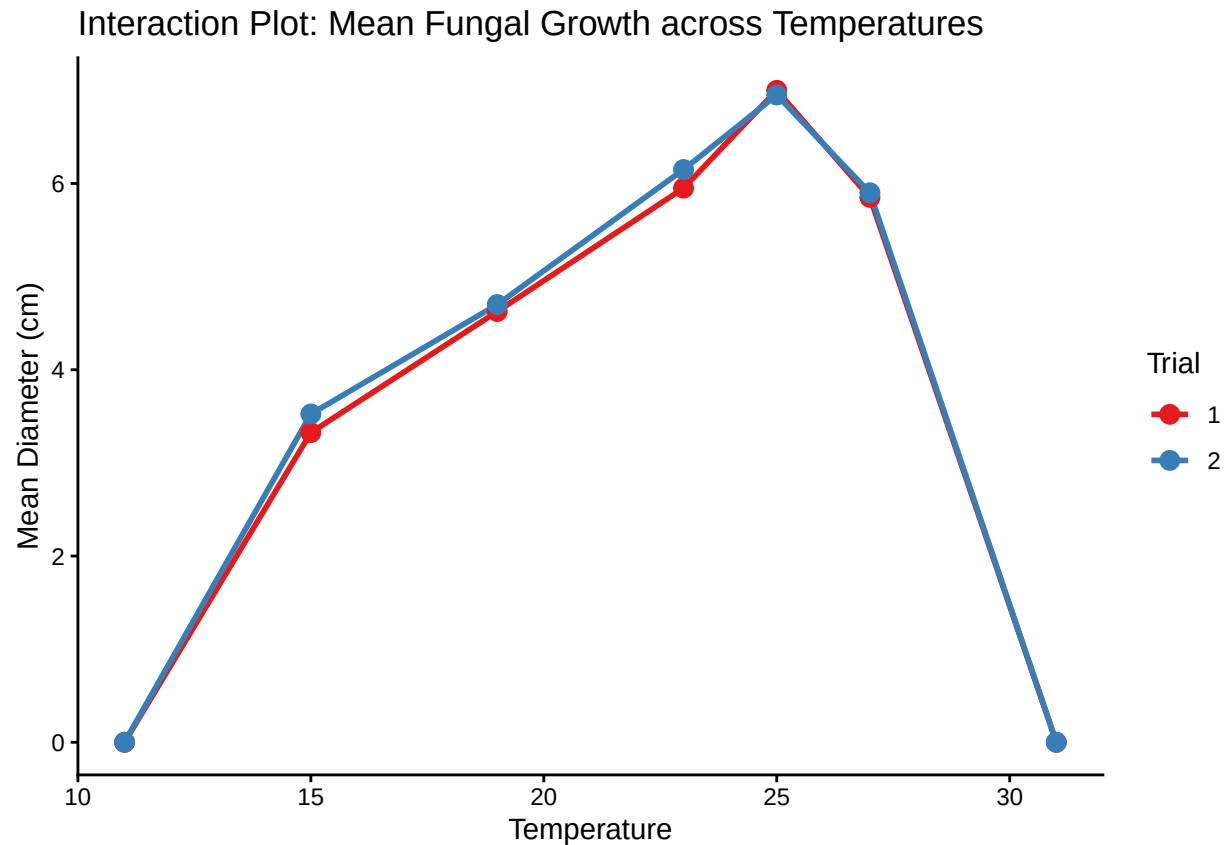
```
head(ph_data)
```

```
##   pH Trial Rep Diameter
## 1 3.0    1  1    0.000
## 2 3.0    1  2    0.000
## 3 3.0    1  3    0.000
## 4 4.5    1  1    0.179
## 5 4.5    1  2    0.089
## 6 4.5    1  3    0.100
```

EDA

```
# Temperature Interaction Plot
ggplot(temp_data, aes(x = as.numeric(as.character(Temp)), y = Diameter,
                      color = as.factor(Trial), group = Trial)) +
  stat_summary(fun = mean, geom = "line", size = 1) +
  stat_summary(fun = mean, geom = "point", size = 3) +
  scale_color_brewer(palette = "Set1") + # This gives you the Red/Blue distinct colors
  labs(title = "Interaction Plot: Mean Fungal Growth across Temperatures",
       x = "Temperature",
       y = "Mean Diameter (cm)",
       color = "Trial") +
  theme_classic()
```

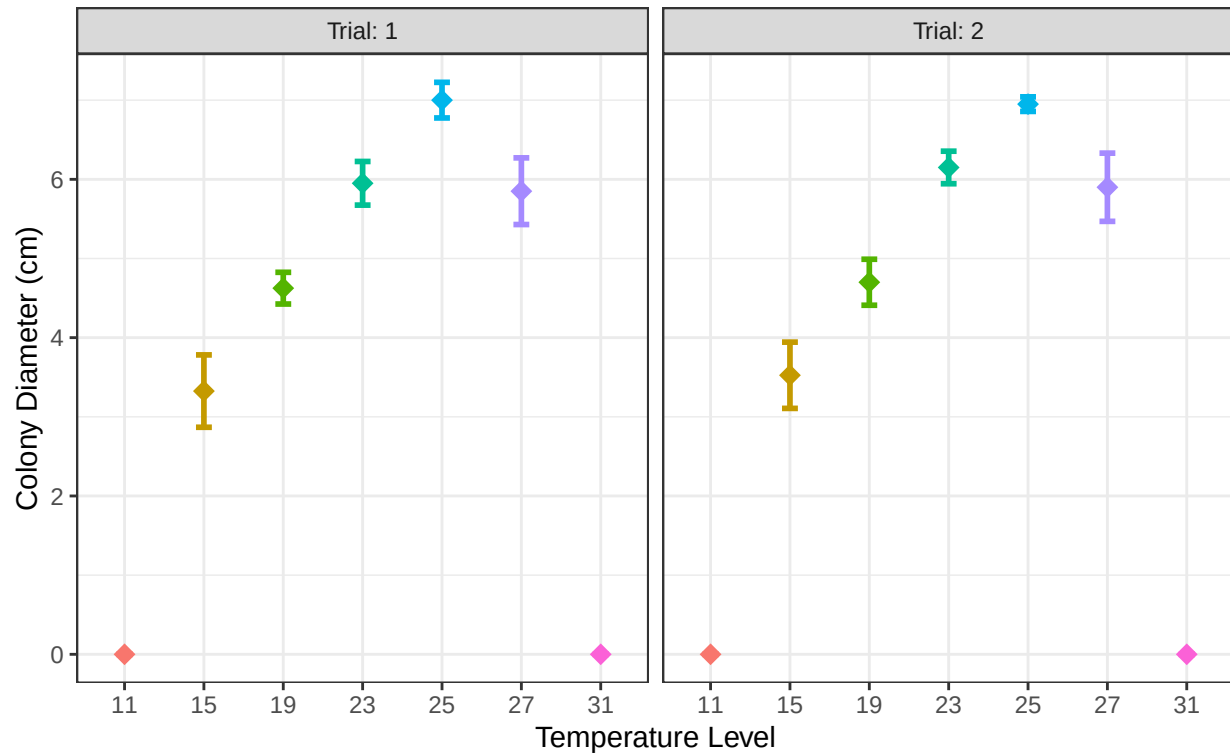
```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



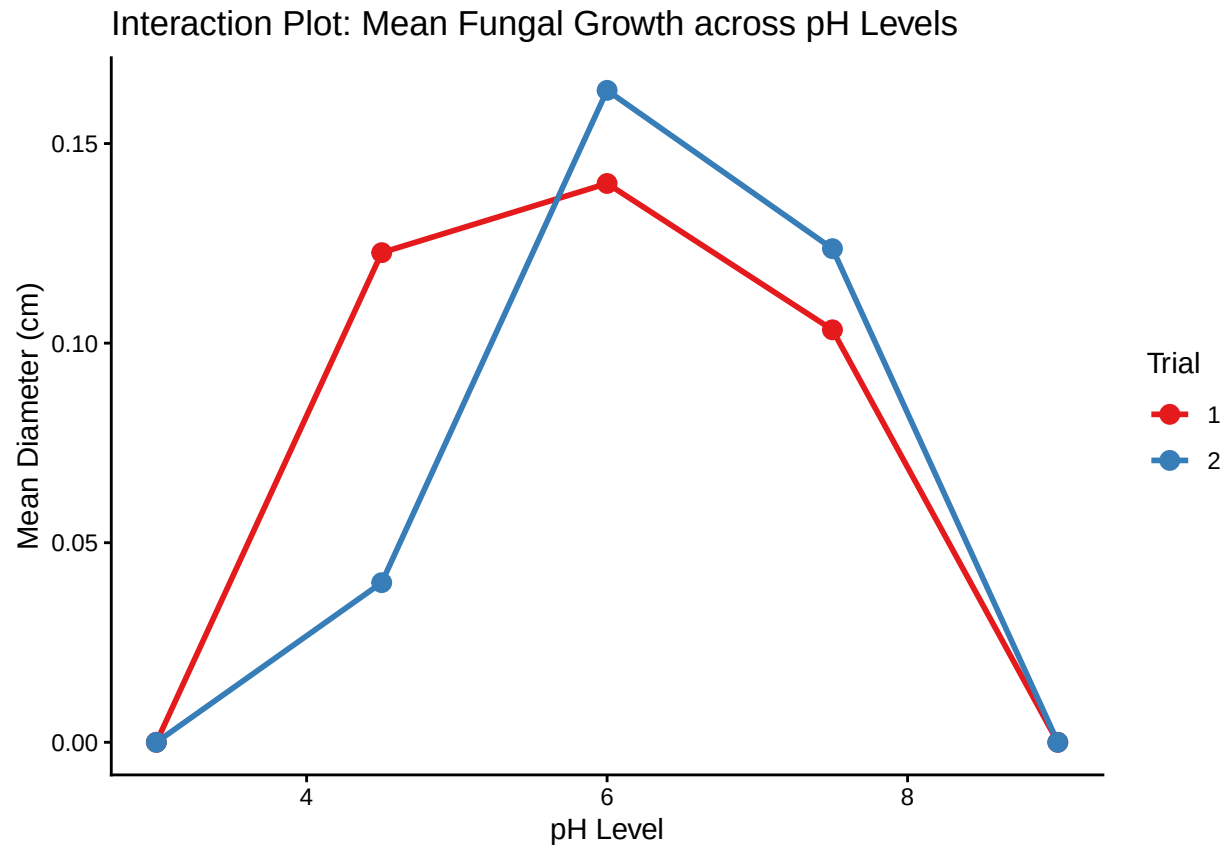
```
# Temperature: Colorful Panels with 95% CI
ggplot(temp_data, aes(x = as.factor(Temp), y = Diameter, color = as.factor(Temp))) +
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2, size = 1) +
  stat_summary(fun = mean, geom = "point", size = 3.5, shape = 18) +
  facet_grid(. ~ Trial, labeller = label_both) +
  labs(
    title = "Mean Fungal Growth (95% CI) Across Temperature Treatments",
    subtitle = "Separated by Experimental Trial to view distribution consistency",
    x = "Temperature Level",
    y = "Colony Diameter (cm)"
  ) +
  theme_bw() +
  theme(legend.position = "none")
```

Mean Fungal Growth (95% CI) Across Temperature Treatments

Separated by Experimental Trial to view distribution consistency

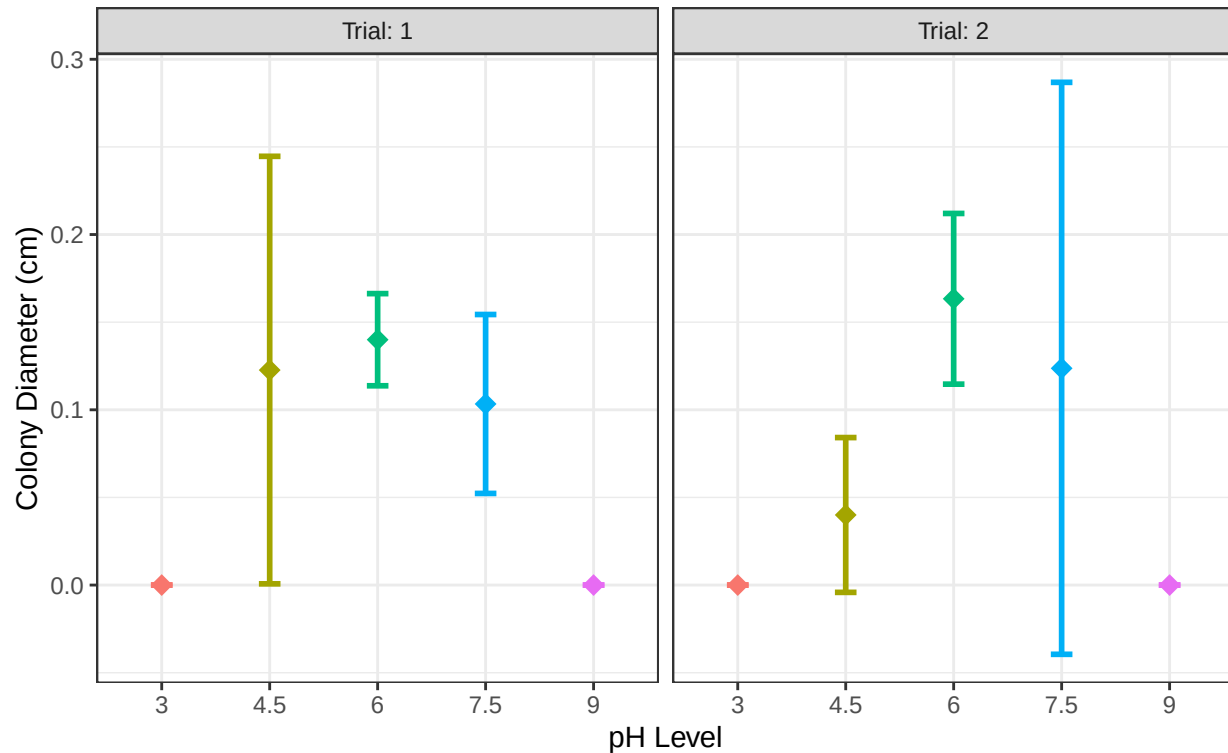


```
# pH Interaction Plot
ggplot(ph_data, aes(x = as.numeric(as.character(pH)), y = Diameter,
                    color = as.factor(Trial), group = Trial)) +
  stat_summary(fun = mean, geom = "line", size = 1) +
  stat_summary(fun = mean, geom = "point", size = 3) +
  scale_color_brewer(palette = "Set1") +
  labs(title = "Interaction Plot: Mean Fungal Growth across pH Levels",
       x = "pH Level",
       y = "Mean Diameter (cm)",
       color = "Trial") +
  theme_classic()
```



```
# pH: Colorful Panels with 95% CI
ggplot(ph_data, aes(x = as.factor(pH), y = Diameter, color = as.factor(pH))) +
  stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2, size = 1) +
  stat_summary(fun = mean, geom = "point", size = 3.5, shape = 18) +
  facet_grid(. ~ Trial, labeller = label_both) +
  labs(
    title = "Mean Fungal Growth (95% CI) Across pH Treatments",
    subtitle = "Separated by Experimental Trial to highlight interaction changes",
    x = "pH Level",
    y = "Colony Diameter (cm)"
  ) +
  theme_bw() +
  theme(legend.position = "none")
```

Mean Fungal Growth (95% CI) Across pH Treatments Separated by Experimental Trial to highlight interaction changes



```
ggplot(temp_data, aes(x = Temp, y = Diameter)) +

  geom_point(size = 2, alpha = 0.7) +

  # Linear
  geom_smooth(
    method = "lm",
    formula = y ~ x,
    aes(linetype = "Linear"),
    se = FALSE
  ) +

  # Quadratic
  geom_smooth(
    method = "lm",
    formula = y ~ poly(x, 2),
    aes(linetype = "Quadratic"),
    se = FALSE
  ) +

  # Cubic
  geom_smooth(
    method = "lm",
    formula = y ~ poly(x, 3),
    aes(linetype = "Cubic"),
    se = FALSE
  )
```

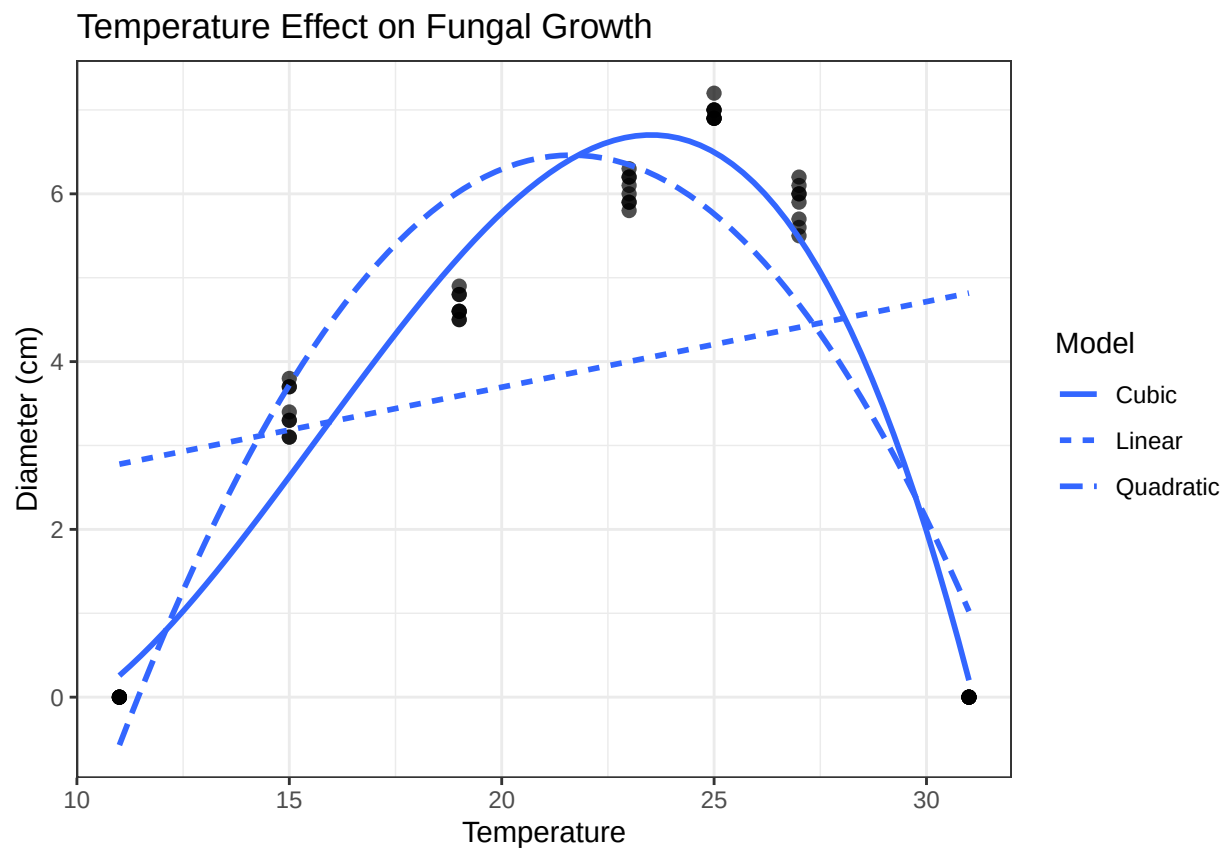
```

) +

labs(
  title = "Temperature Effect on Fungal Growth",
  x = "Temperature",
  y = "Diameter (cm)",
  linetype = "Model"
) +

theme_bw()

```



```

ggplot(ph_data, aes(x = pH, y = Diameter)) +

  geom_point(size = 2, alpha = 0.7) +

  geom_smooth(
    method = "lm",
    formula = y ~ x,
    aes(linetype = "Linear"),
    se = FALSE
  ) +

  geom_smooth(
    method = "lm",
    formula = y ~ poly(x, 2),

```

```

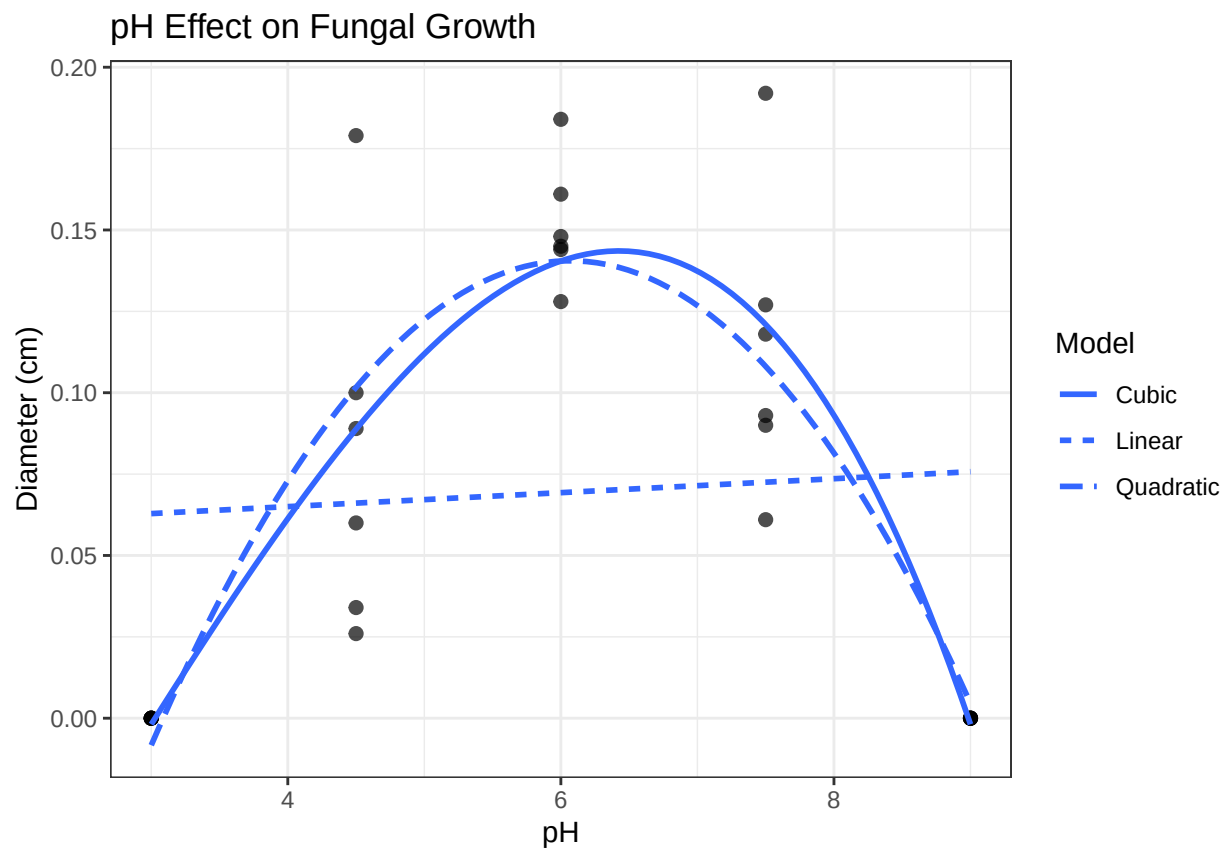
aes(linetype = "Quadratic"),
se = FALSE
) +

geom_smooth(
method = "lm",
formula = y ~ poly(x, 3),
aes(linetype = "Cubic"),
se = FALSE
) +

labs(
title = "pH Effect on Fungal Growth",
x = "pH",
y = "Diameter (cm)",
linetype = "Model"
) +

theme_bw()

```



```

## Filter for rows where Diameter is NOT equal to 0
temp_data <- temp_data %>% filter(Diameter != 0)
ph_data    <- ph_data    %>% filter(Diameter != 0)

```



```

# 1. Fit the three nested models
temp_linear <- lm(Diameter ~ Temp, data = temp_data)
temp_quad   <- lm(Diameter ~ Temp + I(Temp^2), data = temp_data)
temp_cubic  <- lm(Diameter ~ (Temp + I(Temp^2) + I(Temp^3)), data = temp_data)

# 2. Perform the ANOVA Likelihood Ratio Test
temp_comparison <- anova(temp_linear, temp_quad, temp_cubic)
print(temp_comparison)

```

```

## Analysis of Variance Table
##
## Model 1: Diameter ~ Temp
## Model 2: Diameter ~ Temp + I(Temp^2)
## Model 3: Diameter ~ (Temp + I(Temp^2) + I(Temp^3))
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      38 13.2402
## 2      37  8.1632  1     5.077 54.313 1.081e-08 ***
## 3      36  3.3652  1     4.798 51.328 2.001e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

# 1. Fit the nested models
ph_linear <- lm(Diameter ~ pH, data = ph_data)
ph_quad   <- lm(Diameter ~ pH + I(pH^2), data = ph_data)

# 2. Perform the ANOVA Likelihood Ratio Test
ph_comparison <- anova(ph_linear, ph_quad)
print(ph_comparison)

```

```

## Analysis of Variance Table
##
## Model 1: Diameter ~ pH
## Model 2: Diameter ~ pH + I(pH^2)
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      16 0.039378
## 2      15 0.027606  1  0.011772 6.3965 0.02313 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Temp model

```

temp_model <- lm(Diameter ~ (Temp + I(Temp^2) + I(Temp^3)) * Trial, data = temp_data)

cat("\n--- Temperature Model Summary ---\n")

```

```

##
## --- Temperature Model Summary ---

```

```
print(summary(temp_model))
```

```
##
## Call:
## lm(formula = Diameter ~ (Temp + I(Temp^2) + I(Temp^3)) * Trial,
##     data = temp_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.52500 -0.24454  0.01142  0.23941  0.60000
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   50.663106   30.582332    1.657   0.1074
## Temp          -7.938701    4.547405   -1.746   0.0904 .
## I(Temp^2)      0.424863    0.220121    1.930   0.0625 .
## I(Temp^3)     -0.007117    0.003480   -2.045   0.0491 *
## Trial           3.036503   19.341965    0.157   0.8762
## Temp:Trial     -0.410127    2.876032   -0.143   0.8875
## I(Temp^2):Trial 0.019277    0.139217    0.138   0.8907
## I(Temp^3):Trial -0.000305    0.002201   -0.139   0.8907
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3182 on 32 degrees of freedom
## Multiple R-squared:  0.9477, Adjusted R-squared:  0.9363
## F-statistic: 82.91 on 7 and 32 DF,  p-value: < 2.2e-16
```

```
cat("\n--- ANOVA Table for Temperature Regression ---\n")
```

```
##
## --- ANOVA Table for Temperature Regression ---
```

```
print(anova(temp_model))
```

```
## Analysis of Variance Table
##
## Response: Diameter
##              Df Sum Sq Mean Sq F value    Pr(>F)
## Temp           1  48.770   48.770  481.6147 < 2.2e-16 ***
## I(Temp^2)       1   5.077    5.077   50.1372 4.956e-08 ***
## I(Temp^3)       1   4.798    4.798   47.3821 8.642e-08 ***
## Trial           1   0.090    0.090    0.8912  0.3522
## Temp:Trial      1   0.033    0.033    0.3220  0.5744
## I(Temp^2):Trial 1   0.000    0.000    0.0000  0.9989
## I(Temp^3):Trial 1   0.002    0.002    0.0192  0.8907
## Residuals      32   3.240    0.101
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
temp_model_final <- lm(Diameter ~ (Temp + I(Temp^2) + I(Temp^3)), data = temp_data)
cat("\n--- Temperature Model Summary ---\n")
```

```
##
## --- Temperature Model Summary ---
```

```
print(summary(temp_model_final))
```

```
##
## Call:
## lm(formula = Diameter ~ (Temp + I(Temp^2) + I(Temp^3)), data = temp_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.57037 -0.19537  0.01821  0.25691  0.56673
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  55.217861    9.291814   5.943 8.29e-07 ***
## Temp        -8.553892    1.381636  -6.191 3.86e-07 ***
## I(Temp^2)     0.453779    0.066879   6.785 6.29e-08 ***
## I(Temp^3)    -0.007574    0.001057  -7.164 2.00e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3057 on 36 degrees of freedom
## Multiple R-squared:  0.9457, Adjusted R-squared:  0.9412
## F-statistic: 209.1 on 3 and 36 DF,  p-value: < 2.2e-16
```

```
cat("\n--- ANOVA Table for Temperature Regression ---\n")
```

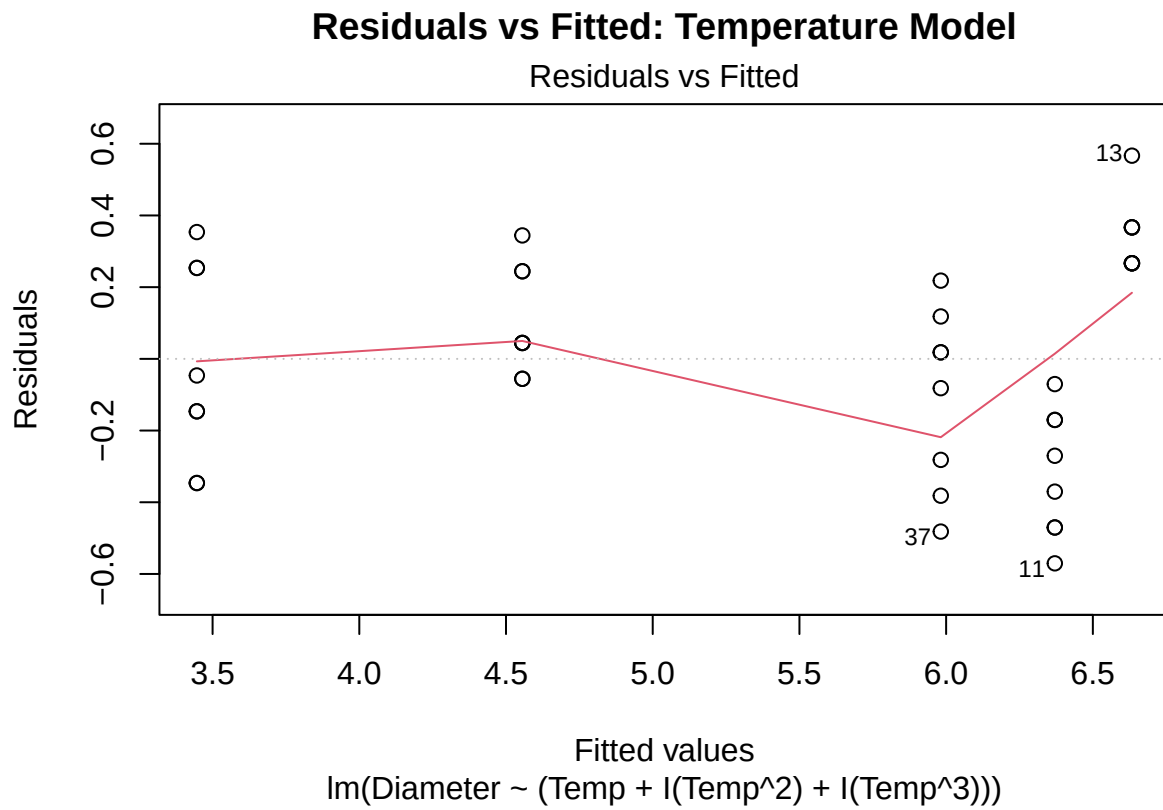
```
##
## --- ANOVA Table for Temperature Regression ---
```

```
print(anova(temp_model_final))
```

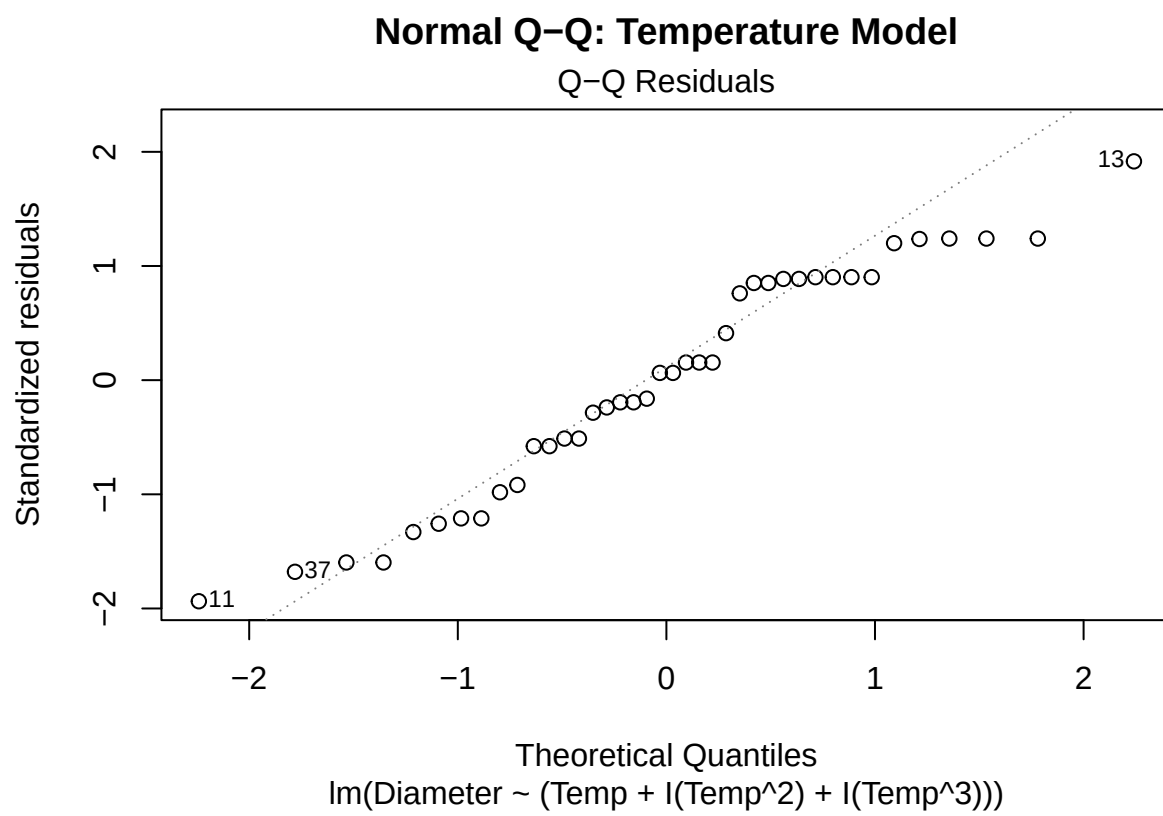
```
## Analysis of Variance Table
##
## Response: Diameter
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Temp       1  48.770   48.770  521.723 < 2.2e-16 ***
## I(Temp^2)   1   5.077    5.077   54.313 1.081e-08 ***
## I(Temp^3)   1   4.798    4.798   51.328 2.001e-08 ***
## Residuals  36   3.365    0.093
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

assumption check

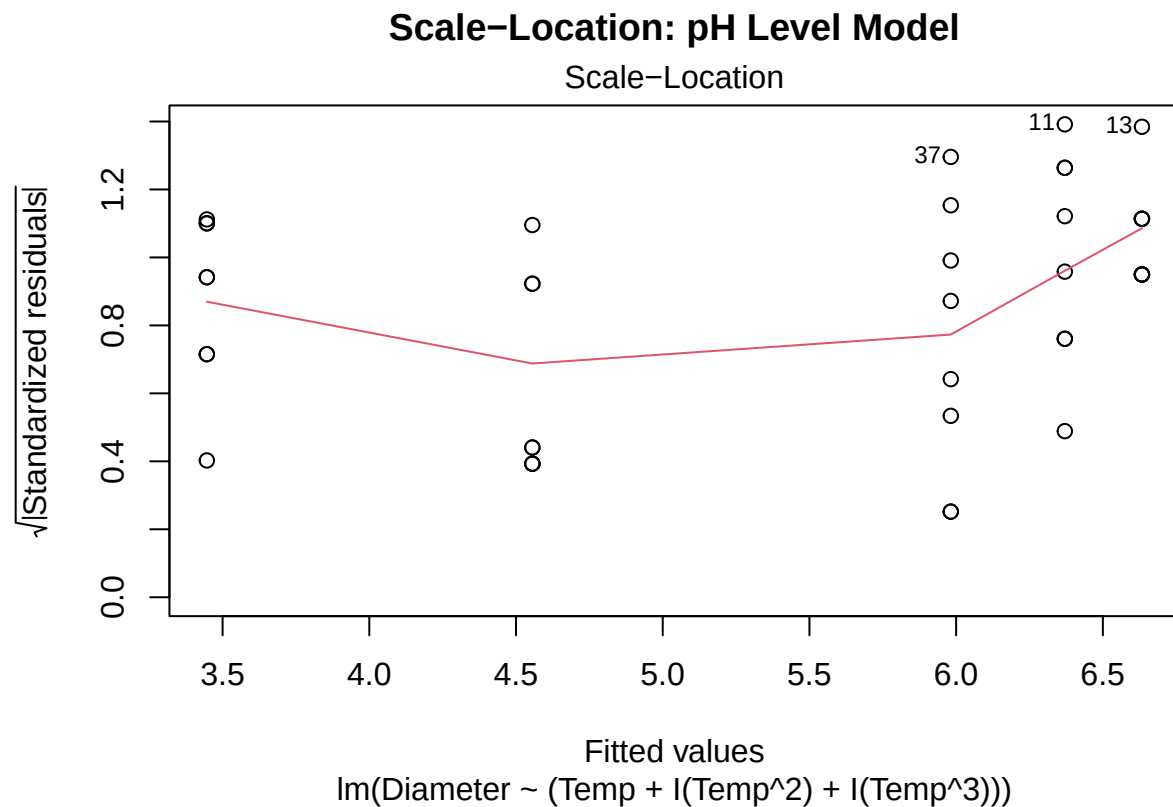
```
plot(temp_model_final, which = 1, main = "Residuals vs Fitted: Temperature Model")
```



```
plot(temp_model_final, which = 2, main = "Normal Q-Q: Temperature Model")
```



```
plot(temp_model_final, which = 3, main = "Scale-Location: pH Level Model")
```



```
durbinWatsonTest(temp_model_final)
```

```
## lag Autocorrelation D-W Statistic p-value
## 1 0.2617965 1.44066 0.026
## Alternative hypothesis: rho != 0
```

```
shapiro.test(residuals(temp_model_final))
```

```
##
## Shapiro-Wilk normality test
##
## data: residuals(temp_model_final)
## W = 0.95472, p-value = 0.1102
```

ph model

```
ph_model <- lm(Diameter ~ (pH + I(pH^2)) * Trial, data = ph_data)
```

```
cat("\n--- pH Model Summary ---\n")
```

```
##
## --- pH Model Summary ---
```

```
print(summary(ph_model))
```

```
##
## Call:
## lm(formula = Diameter ~ (pH + I(pH^2)) * Trial, data = ph_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.062667 -0.013833 -0.005833  0.017000  0.068333
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.80133    0.89339   0.897   0.3874
## pH            -0.18744    0.30794  -0.609   0.5541
## I(pH^2)        0.01222    0.02559   0.478   0.6416
## Trial          -1.05467    0.56503  -1.867   0.0866 .
## pH:Trial       0.32500    0.19476   1.669   0.1210
## I(pH^2):Trial -0.02422    0.01619  -1.496   0.1604
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03642 on 12 degrees of freedom
## Multiple R-squared:  0.6253, Adjusted R-squared:  0.4692
## F-statistic: 4.005 on 5 and 12 DF,  p-value: 0.02274
```

```
cat("\n--- ANOVA Table for pH Regression ---\n")
```

```
##
## --- ANOVA Table for pH Regression ---
```

```
print(anova(ph_model))
```

```
## Analysis of Variance Table
##
## Response: Diameter
##      Df    Sum Sq  Mean Sq F value Pr(>F)
## pH      1 0.0031041 0.0031041  2.3400 0.15202
## I(pH^2)  1 0.0117723 0.0117723  8.8743 0.01151 *
## Trial    1 0.0007605 0.0007605  0.5733 0.46356
## pH:Trial 1 0.0079568 0.0079568  5.9981 0.03064 *
## I(pH^2):Trial 1 0.0029703 0.0029703  2.2391 0.16039
## Residuals 12 0.0159187 0.0013266
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ph_model_final <- lm(Diameter ~ (pH) * Trial + I(pH^2), data = ph_data)
```

```
cat("\n--- pH Model Summary ---\n")
```

```
##
## --- pH Model Summary ---
```

```
print(summary(ph_model_final))
```

```
##
## Call:
## lm(formula = Diameter ~ (pH) * Trial + I(pH^2), data = ph_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.071750 -0.015000 -0.007208  0.014604  0.065417
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.452167   0.324937  -1.392   0.1874
## pH           0.248556   0.104262   2.384   0.0331 *
## Trial        -0.219000   0.089845  -2.438   0.0299 *
## I(pH^2)      -0.024111   0.008471  -2.846   0.0138 *
## pH:Trial     0.034333   0.014672   2.340   0.0359 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03812 on 13 degrees of freedom
## Multiple R-squared:  0.5554, Adjusted R-squared:  0.4186
## F-statistic: 4.059 on 4 and 13 DF,  p-value: 0.02375
```

```
cat("\n--- ANOVA Table for pH Regression ---\n")
```

```
##
## --- ANOVA Table for pH Regression ---
```

```
print(anova(ph_model_final))
```

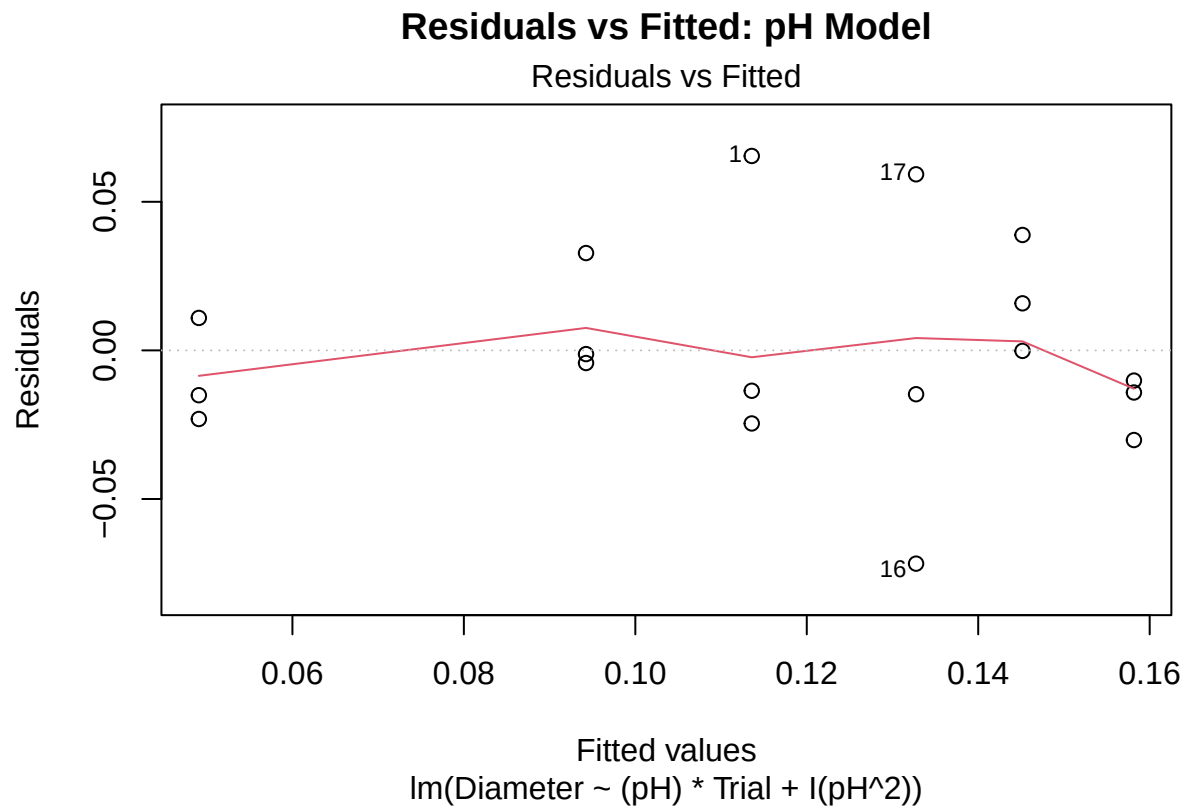
```
## Analysis of Variance Table
##
## Response: Diameter
##      Df    Sum Sq  Mean Sq F value  Pr(>F)
## pH      1 0.0031041 0.0031041   2.1363 0.16759
## Trial    1 0.0007605 0.0007605   0.5234 0.48221
## I(pH^2)  1 0.0117723 0.0117723   8.1021 0.01375 *
## pH:Trial  1 0.0079568 0.0079568   5.4761 0.03588 *
## Residuals 13 0.0188889 0.0014530
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
trial_comparison <- anova(ph_model_final, ph_model)
print(trial_comparison)
```

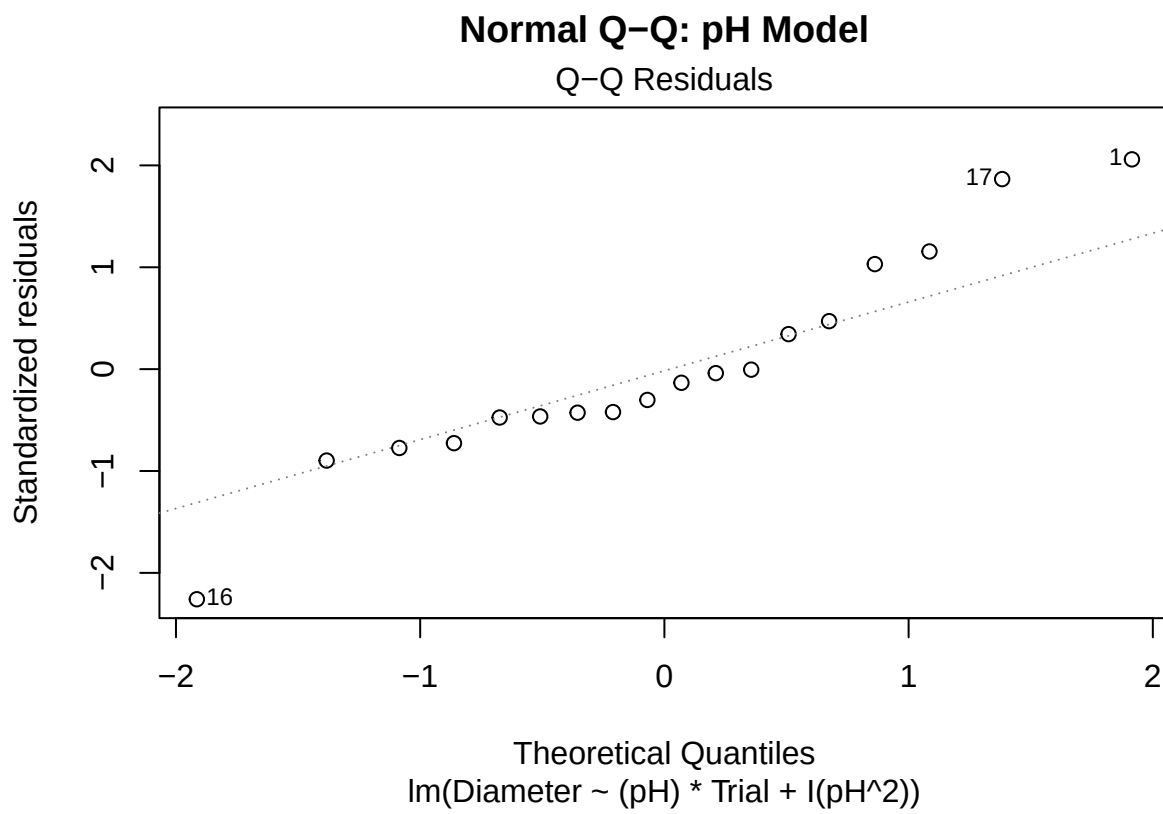
```
## Analysis of Variance Table
##
## Model 1: Diameter ~ (pH) * Trial + I(pH^2)
## Model 2: Diameter ~ (pH + I(pH^2)) * Trial
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      13 0.018889
## 2      12 0.015919  1 0.0029702 2.2391 0.1604
```


assumption check

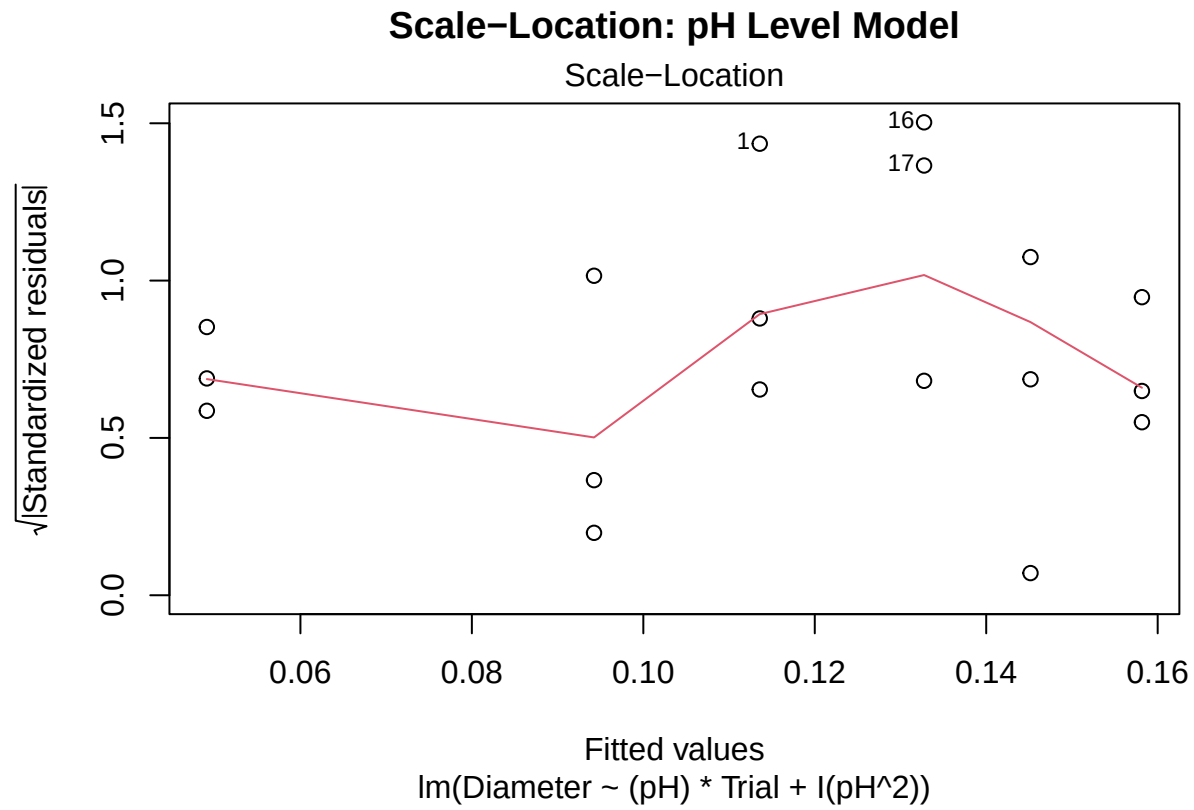
```
plot(ph_model_final, which = 1, main = "Residuals vs Fitted: pH Model")
```



```
plot(ph_model_final, which = 2, main = "Normal Q-Q: pH Model")
```



```
plot(ph_model_final, which = 3, main = "Scale-Location: pH Level Model")
```



```
durbinWatsonTest(ph_model_final)
```

```
## lag Autocorrelation D-W Statistic p-value
## 1 -0.3663822 2.494693 0.904
## Alternative hypothesis: rho != 0
```

```
shapiro.test(residuals(ph_model_final))
```

```
##
## Shapiro-Wilk normality test
##
## data: residuals(ph_model_final)
## W = 0.94305, p-value = 0.3264
```